Interactive Website Implementation and User Evidence

1. Purpose of the Website

Abstract Algebra Lab is an interactive educational implementation supporting the mathematical investigation of three reversible transformation systems: Caesar shifts, invertible affine transformations modulo 26, and fixed-block position permutations from S_4.

The website is not presented as a new cryptographic algorithm or secure messaging system. Its purpose is to make abstract algebraic ideas observable through interaction. Users can select transformations, enter messages, view outputs, inspect inverse operations, and compare the structural properties of the three systems.

The website was deployed through Firebase Hosting at:

https://abstract-algebra-lab.web.app: https://abstract-algebra-lab.web.app

The implementation corresponding to the tested evidence was committed to GitHub in commit `91fb67b13a47bf62ca2e7dedc348ad69a6863e66`.

2. Page-to-Mathematics Structure

The website is organised into seven major views.

The Overview page introduces the research question and presents the three transformation systems:

- Caesar shifts C_26, of order 26.
- Affine transformations AGL(1,26), of order 312.
- Position permutations S_4, of order 24.

The Foundations page explains the group-theoretic concepts needed for the project. It introduces closure, associativity, identity, and inverses, and connects these properties with reversible transformations. It also demonstrates modular arithmetic through a Caesar example such as

E_3(7)=(7+3)mod 26=10.

The Group Explorer page allows users to compare the systems through their operations, orders, composition, and inverses. This supports the paper’s discussion of structural similarities and differences.

The Lab page is the main interactive transformation interface. It provides controls for Caesar, affine, and S■ transformations. Users can enter plaintext, select parameters, encrypt, decrypt, and inspect calculation traces.

The Algebraic Structure page explains the common mathematical framework of the systems: each consists of reversible transformations closed under composition and possessing identity and inverse transformations. It also distinguishes algebraic reversibility from cryptographic security.

The Experiments page contains frequency analysis, transformation inspection, composition and commutativity experiments, and brute-force exploration. These activities demonstrate how algebraic properties influence observable transformation behaviour.

The Research Log records the research question, verified results, methodology, and limitations. It helps separate established mathematical claims from implementation-specific observations.

3. Browser-Verified Transformation Behaviour

The deployed Lab page was tested with all three systems.

For Caesar shifts, the default example used a shift of +3. The plaintext

HELLO, ALGEBRA!

was transformed into

KHOOR, DOJHEUD!.

The inverse control restored the original plaintext. This demonstrates the relationship between a transformation and its inverse in C_26.

The affine interface displayed the valid multiplier values

1,3,5,7,9,11,15,17,19,21,23,25,

which are precisely the residues relatively prime to 26. The interface also provided a shift parameter and displayed the affine group order 312. Invalid multiplier rejection was mathematically enforced by the restricted control options, although direct browser testing of an invalid multiplier was not possible because invalid values were not selectable through the visible interface.

For S_4, the interface exposed all 24 position permutations. The tested example used plaintext

ABCD

with permutation

[2,1,4,3].

Encryption produced

ABCD\longmapstoBADC.

The inverse operation then recovered

BADC\longmapstoABCD.

The calculation trace visibly displayed the ciphertext, inverse permutation, recovered plaintext, and the identity relationship

f^-1(f(x))=x.

This provides direct browser evidence that the implemented S■ workflow demonstrates a complete encryption–decryption round trip.

4. Team-Page Interaction and Accessibility

The Team page presents five project roles and their responsibilities. Profile cards use click-based expansion rather than hover-based content disclosure.

The following behaviours were verified:

- Clicking a profile opens its details.
- Clicking the open profile collapses it.
- Opening another profile closes the previous one.
- Pressing Enter activates a profile.
- Pressing Space activates a profile.
- The expanded profile exposes `aria-expanded="true"`.
- The corresponding `aria-controls` value identifies the correct details region.
- The GitHub call-to-action links to the project repository.

The implementation therefore supports both pointer and keyboard interaction while maintaining a single-open-profile model. Hover styling is treated as visual feedback rather than as a mechanism for revealing content.

5. Dynamic Letter-Mapping Visualisation

An earlier version of the letter-mapping visualisation displayed at most eight mapping cards. This caused additional distinct letters to disappear when a message contained nine or more unique alphabetic characters.

The issue was corrected in the visualisation component. The current implementation:

- Renders every distinct alphabetic character.
- Preserves first-appearance order.
- Handles repeated letters without duplicating cards.
- Allows the mapping tray to wrap across multiple rows.
- Avoids truncating mappings after the eighth card.

The correction was committed and deployed in GitHub commit `91fb67b13a47bf62ca2e7dedc348ad69a6863e66`. The corresponding build and verification checks passed.

6. Verification and Evidence Classification

The project used multiple forms of evidence.

Mathematical claims were established through the approved mathematical specification and independent Python verification.

Software claims were checked using:

npm run lint

npm run verify

npm run build

All three commands passed after the latest changes. The verification script reported:

All Caesar, affine, S4, and frequency verification checks passed successfully.

Browser evidence was collected from the deployed Firebase site. It included page navigation, Caesar and S■ examples, Team-page interaction, accessibility states, console inspection, and the dynamic mapping correction.

The project distinguishes between:

• Mathematical facts proved independently.

• Results verified by automated code.

• Features visible in source code.

• Behaviour observed directly in the browser.

• Features or edge cases not yet tested.

This distinction is important because a feature's presence in source code does not by itself prove that it behaves correctly in the deployed application.

7. Limitations

The website is an educational demonstration of reversible transformations, not a secure cryptographic service. The classical systems studied are useful for illustrating group theory but are not suitable for protecting real confidential communication.

Some browser evidence remains limited to the tested environment and viewport. Image-fallback behaviour and exhaustive responsive breakpoint testing were not treated as fully verified. Similarly, invalid affine input could not be manually entered through the visible restricted selector, although the mathematical engine and automated verification remained successful.

The website also does not replace formal mathematical proof. Its visualisations provide examples and evidence, but conclusions about group structure, element order, cyclicity, subgroups, and isomorphism must be justified mathematically in the research paper.

8. Contribution to the Research

The website contributes an interactive layer to the mathematical investigation. It allows users to move from definitions to observable examples:

abstract operation

->

transformation

->

composition

->

inverse

->

verified output.

Its principal contribution is therefore pedagogical and demonstrative. It shows how abstract algebra provides a common language for analysing reversible transformation systems while also making clear that algebraic structure alone does not establish cryptographic security.